

Applications of Statistical Modeling:

Module #2, Lecture #3

Applying LCA to Data from the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC)

Instructor: Brady T. West
Email: bwest@umich.edu

Nonmedical Use of Prescription Medication (NMUPM)

- Nonmedical use of prescription medications has become a critical public health problem in the United States, with serious health implications (addiction, illness, co-ingestion)
- **Brief Definition:** The use of prescription medications not actually prescribed to a person by a doctor, or in a manner not intended by the prescribing clinician

Research Question

- The longitudinal nature of the NESARC design allows us to address causal questions!
- **Primary Research Question:** Does past-year NMUPM in 2001-2002 predict the probability of developing substance use disorders three years later, in 2004-2005?

Measurement Error?

- NESARC respondents indicate whether they have engaged in NMUPM to a live human interviewer in the FTF interviews
- Because this is a sensitive behavior, these reports may be subject to measurement error
- We will use LCA to identify possible classes of individuals based on (error-prone?) indicators of past-year nonmedical use of 4 prescription medications, and heavy drinking (in '01-'02)

Measurement Error?

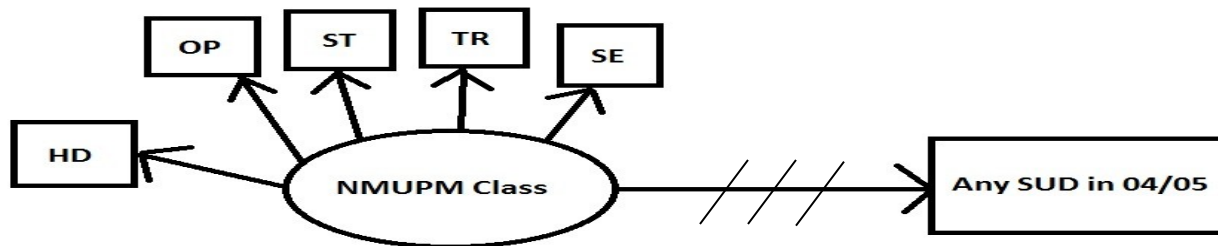
- The four prescription medication classes:
 - Opioids / Analgesics
 - Stimulants
 - Tranquilizers
 - Sedatives
- Allowing for possible measurement error, we will use these four indicators and the indicator of heavy drinking to determine classes of individuals based on their reported behaviors

Analytic Approach

- Identify and profile classes of individuals (LCA)
- In a separate step, use the predicted latent classes in 2001-2002 to predict the probability of developing any type of substance use disorder in 2004-2005
- For details regarding this type of multi-step approach to using latent classes for predicting a **distal outcome**, see Asparouhov and Muthen (2014a, 2014b) on Canvas

Overall SEM

- Note how this model combines a latent class analysis (LCA), for **measurement**, and a logistic regression model, for **structure**
- We need a multi-step approach to the analysis (LCA, *then* predicting the **distal outcome**); otherwise, our DV is just another indicator!



Analytic Approach

- Ultimately, do the identified classes in 2001-2002 vary in terms of the probability of developing any SUD in 2004-2005?
- We will begin with simple descriptive statistics, continue with the LCA (including profiles), and then fit the overall model
- Next class, we will consider how to account for the complex sample design features of the NESARC when performing the analysis

The NESARC Data Set

- Available on Canvas (nesarc_w1w2_red.XXX)
- $n = 34,653$
- Variables of interest for this illustration:
 - HD: Heavy drinking in past year (01-02)
 - ANYSEDY: Any NM use of sedatives in past year
 - ANYTRQY: Any NM use of tranquilizers in past year
 - ANYOPIY: Any NM use of opioids in past year
 - ANYSTMY: Any NM use of stimulants in past year
 - ANYDUDA: Any drug use disorder in past year (04-05); diagnosis based on a series of survey questions related to alcohol, prescription drugs, and illicit drugs

Descriptive Statistics

- Normally, we should be computing weighted estimates (to describe the population) and design-based standard errors for the estimates; we will revisit this next class!
- For now, we will simply compute descriptive statistics for the sample, to get a sense of the data with which we are working

Descriptive Statistics

- Here are the percentages of cases having a 1 on each observed indicator variable (note that there is some missing data!):
 - Any Sedative Use: $397 / 34,643 = 1.15\%$
 - Any Tranquilizer Use: $273 / 34,652 = 0.79\%$
 - Any Opioid Use: $547 / 34,649 = 1.58\%$
 - Any Stimulant Use: $151 / 34,652 = 0.44\%$
 - Heavy Drinking: $7,960 / 34,540 = 23.05\%$

Latent Class Analysis (Mplus)

- Now, let's perform the Latent Class Analysis (LCA) of these data in Mplus
- We choose Mplus because it provides “automatic” procedures for implementing the theoretically correct multi-step analysis; this approach is not readily implemented in other software
- Estimation of the model using “standard” LCA, treating the distal outcome like any other indicator, could lead to biased estimates of the relationships of interest

Latent Class Analysis (Mplus)

- Start → All Programs → Advanced Statistics → Mplus X.X → Mplus Editor
- File → New (Open New Program Editor)
- Enter the syntax on the following slide
- Save the program as a .inp (input) file
- Mplus → Run Mplus (or Alt + R); **any errors?**
- Examine and Interpret Output (.out file)
- Graphs → View Graphs → Select Graph

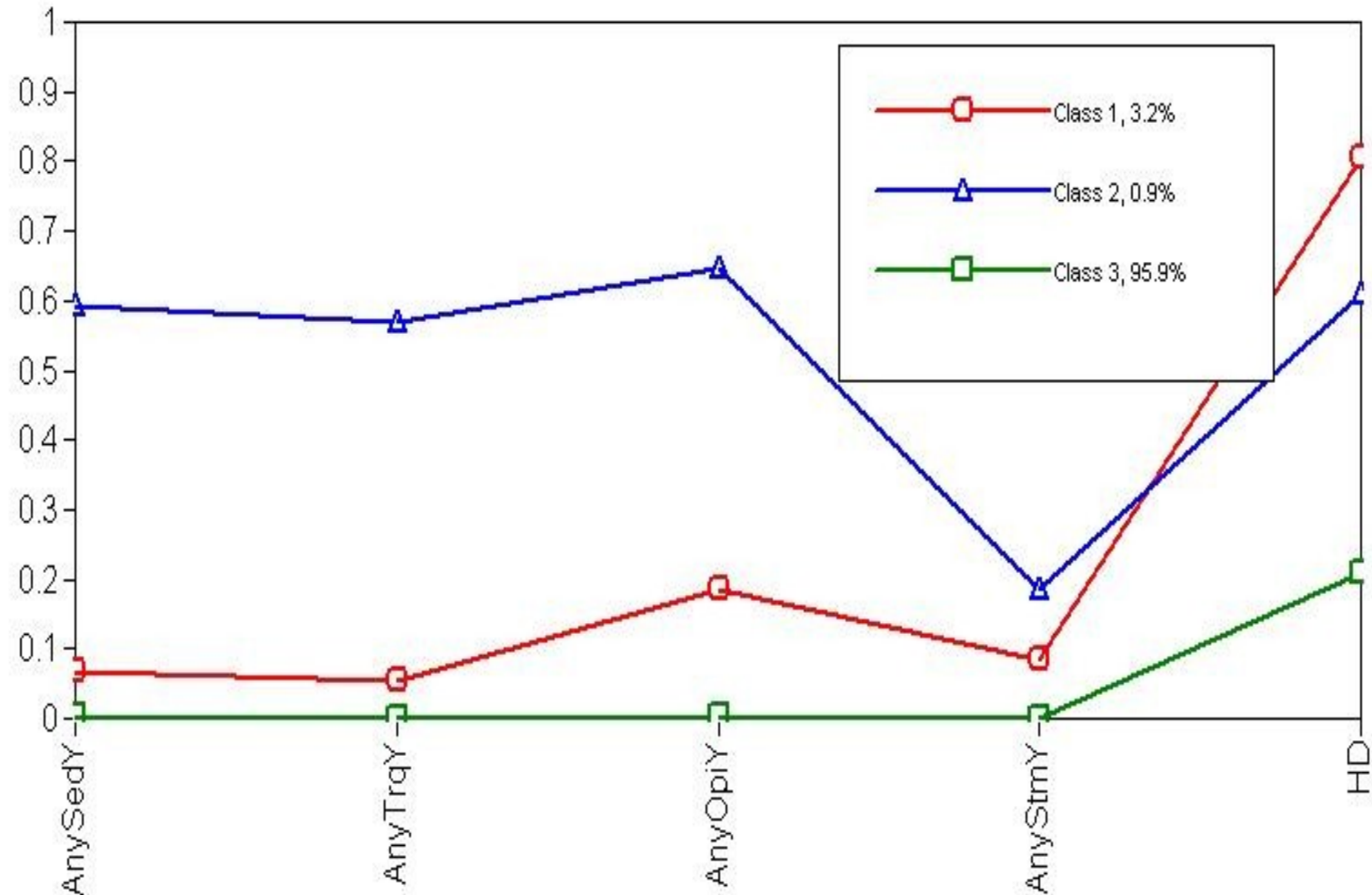
Syntax for Initial NESARC LCA

```
TITLE:          NESARC Latent Class Analysis Example
DATA:           FILE = M:\nesarc_w1w2_red.dat;
VARIABLE: NAMES = idnum anysedy anytrqy anyopiy anystmy hd
               anyduda;
               AUXILIARY = idnum anyduda; ! for saved data file
               USEVARIABLES = anysedy anytrqy anyopiy anystmy
hd;
               CATEGORICAL = anysedy anytrqy anyopiy anystmy
hd;
               MISSING = anysedy (999) anytrqy (999) anyopiy
(999) anystmy (999) hd (999);
               CLASSES = c(3);
ANALYSIS: TYPE = MIXTURE;
PLOT:           TYPE = PLOT3;
               SERIES = anysedy anytrqy anyopiy anystmy hd(*);
OUTPUT:        TECH11 TECH14;
SAVEDATA: FILE IS M:\cprobs.dat;
               SAVE IS CPROB; ! save predicted class membership
```

LCA Output

- Always best to start with plots of the LCA solution first; the Mplus output can be extensive and difficult to process
- Default graphs can be edited; you'll usually want to edit the X-axis (Graphs → View Graphs → Estimated Probabilities, Right-click on X-axis, Axis Properties, Edit)
- These graphs enable you to **profile** the three estimated latent classes

Initial Plot of Three-Class Solution



Initial Plot of Three-Class Solution

- We see evidence of three distinct classes, with Class 3 being the most common class
- Class 3 (green) seems to be the “non-users” who occasionally drink heavily
- **Class 1 (red)** seems to be the “occasional users” who dabble in NMUPM and drink heavily (highest estimated probability)
- **Class 2 (blue)** are the rare problematic cases with high probabilities of NMUPM **and** HD

Three-Class Solution?

- This may not be the only solution! What about four classes? Five classes? Let's take a closer look at the output...

Model Fit: Three-Class Solution

Information Criteria

Akaike (AIC)	49852.105
Bayesian (BIC)	49995.809
Sample-Size Adjusted BIC	49941.783
$(n^* = (n + 2) / 24)$	

Chi-Square Test of Model Fit for the Binary and Ordered Categorical (Ordinal) Outcomes**

Pearson Chi-Square

Value	32.784
Degrees of Freedom	13
P-Value	0.0018

Likelihood Ratio Chi-Square

Value	37.433
Degrees of Freedom	13
P-Value	0.0004

Estimated Class Probabilities

FINAL CLASS COUNTS AND PROPORTIONS FOR THE LATENT CLASS PATTERNS
BASED ON ESTIMATED POSTERIOR PROBABILITIES

Latent
Classes

1	1102.21969	0.03181
2	306.55037	0.00885
3	33244.22994	0.95935

CLASSIFICATION QUALITY

Entropy **0.904**

Predicting Individual Classes

CLASSIFICATION OF INDIVIDUALS **BASED ON THEIR MOST LIKELY LATENT CLASS MEMBERSHIP**

Class Counts and Proportions

Latent

Classes

1	429	0.01238
2	232	0.00669
3	33992	0.98093

Average Latent Class Probabilities for Most Likely Latent Class Membership (Row) by Latent Class (Column)

	1	2	3
1	0.701	0.140	0.159
2	0.121	0.874	0.005
3	0.023	0.001	0.976

Estimated Indicator Probabilities

RESULTS IN PROBABILITY SCALE

Latent Class 1

ANYSEDY

Category 1	0.932	0.024	38.410	0.000
Category 2	0.068	0.024	2.812	0.005

ANYTRQY

Category 1	0.947	0.028	34.266	0.000
Category 2	0.053	0.028	1.930	0.054

ANYOPIY

Category 1	0.813	0.059	13.725	0.000
Category 2	0.187	0.059	3.154	0.002

ANYSTMY

Category 1	0.914	0.029	31.820	0.000
Category 2	0.086	0.029	2.979	0.003

HD

Category 1	0.194	0.050	3.903	0.000
Category 2	0.806	0.050	16.232	0.000

VLMR Likelihood Ratio Test (TECH11)

VUONG-LO-MENDELL-RUBIN LIKELIHOOD RATIO TEST FOR 2 (H0) VERSUS 3 CLASSES

H0 Loglikelihood Value	-24950.796
2 Times the Loglikelihood Difference	83.488
Difference in the Number of Parameters	6
Mean	4.891
Standard Deviation	4.697
P-Value	0.0000

LO-MENDELL-RUBIN ADJUSTED LRT TEST

Value	82.177
P-Value	0.0000

We strongly reject the null hypothesis of a two-class solution.

Bootstrap Likelihood Ratio Test (TECH14)

PARAMETRIC BOOTSTRAPPED LIKELIHOOD RATIO TEST FOR 2 (H0) VERSUS 3 CLASSES

H0 Loglikelihood Value	-24950.796
2 Times the Loglikelihood Difference	83.488
Difference in the Number of Parameters	6
Approximate P-Value	0.0000
Successful Bootstrap Draws	10

We reach the same result here.

Output Data File

SAVEDATA INFORMATION

Order and format of variables

ANYSEDY	F10.3
ANYTRQY	F10.3
ANYOPIY	F10.3
ANYSTMY	F10.3
HD	F10.3
IDNUM	F10.3
ANYDUDA	F10.3
CPROB1	F10.3
CPROB2	F10.3
CPROB3	F10.3
C	F10.3

Save file

M:\cprobs.dat **(USEFUL FOR ADDITIONAL ANALYSIS AND EXPLORATION!)**

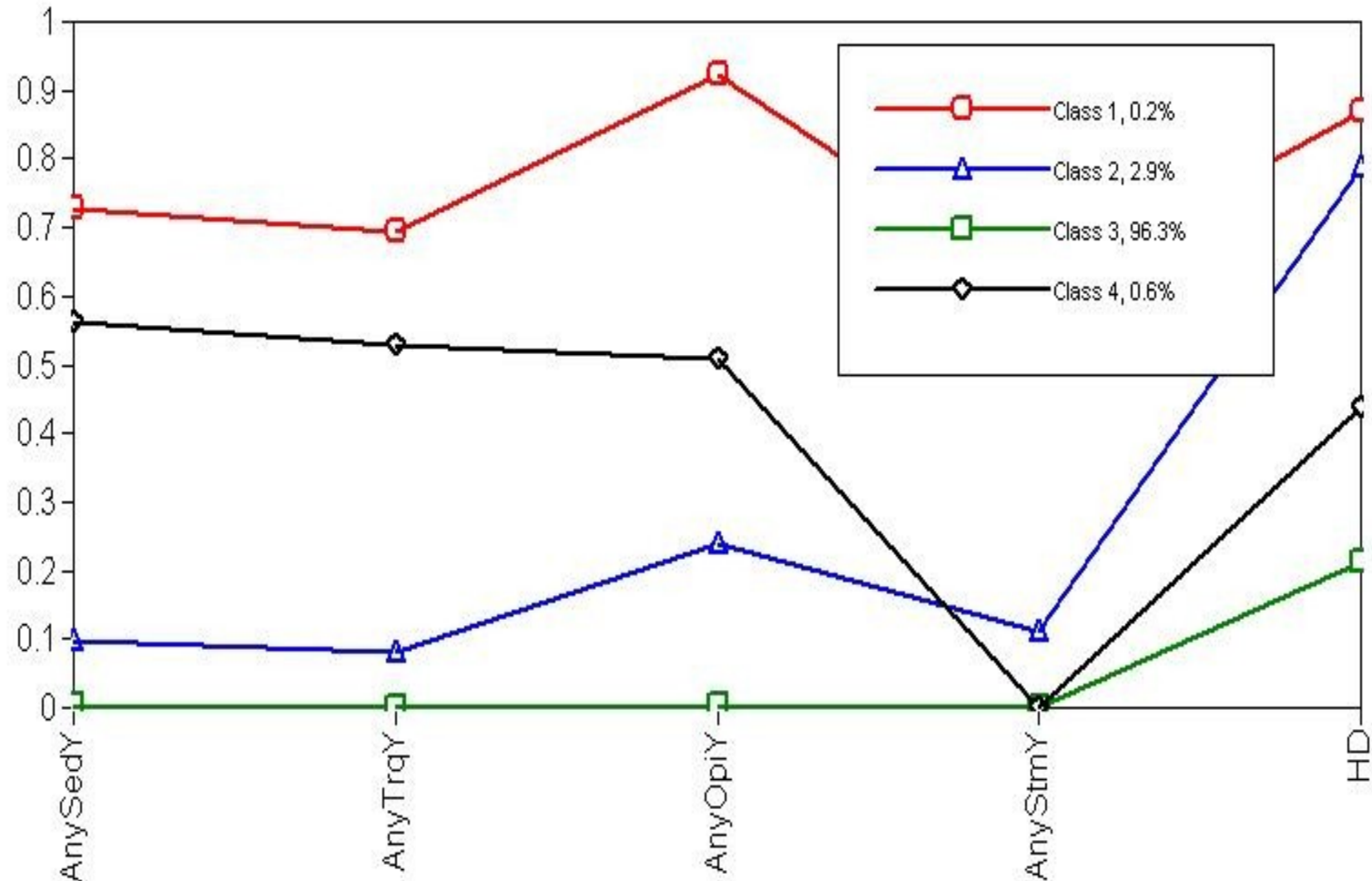
What About Four Classes?

- Now, we consider a four-class solution
- Only one part of the previous syntax needs to be modified in this case:

```
CLASSES = c(4) ;
```

- After re-running this syntax, we re-consider a plot of the solution, along with the model fit statistics and likelihood ratio tests
- **Note:** this will take longer to run!

Plot of Four-Class Solution



Plot of Four-Class Solution

- We see that the four-class solution splits the most problematic group into even smaller groups (**red** and **black**), which are distinguished in terms of probability of use
- **Class 1** appears to be particularly problematic
- Keeping future analysis in mind, this may or may not be ideal; one group may be too small, but could still warrant additional investigation

Model Fit: Four-Class Solution

Information Criteria

Akaike (AIC)	49836.896
--------------	-----------

Bayesian (BIC)	50031.318
-----------------------	------------------

Sample-Size Adjusted BIC	49958.224
--------------------------	-----------

$(n^* = (n + 2) / 24)$

Chi-Square Test of Model Fit for the Binary and Ordered Categorical (Ordinal) Outcomes**

Pearson Chi-Square

Value	7.105
-------	-------

Degrees of Freedom	7
--------------------	---

P-Value	0.4181
----------------	---------------

Likelihood Ratio Chi-Square

Value	10.269
-------	--------

Degrees of Freedom	7
--------------------	---

P-Value	0.1739
----------------	---------------

Evidence of a better fit, although the BIC is slightly larger!

VLMR Likelihood Ratio Test (TECH11)

VUONG-LO-MENDELL-RUBIN LIKELIHOOD RATIO TEST FOR 3 (H0) VERSUS 4 CLASSES

H0 Loglikelihood Value	-24908.984
2 Times the Loglikelihood Difference	27.073
Difference in the Number of Parameters	6
Mean	7.624
Standard Deviation	10.463
P-Value	0.0511

LO-MENDELL-RUBIN ADJUSTED LRT TEST

Value	26.648
P-Value	0.0532

We would fail to reject the null hypothesis of a three-class solution (using $p = 0.05$ strictly!).

Bootstrap Likelihood Ratio Test (TECH14)

PARAMETRIC BOOTSTRAPPED LIKELIHOOD RATIO TEST FOR 3 (H0) VERSUS 4 CLASSES

H0 Loglikelihood Value	-24908.984
2 Times the Loglikelihood Difference	27.073
Difference in the Number of Parameters	6
Approximate P-Value	0.0000
Successful Bootstrap Draws	20

We see more evidence against the null here...slightly mixed results!

Four-Class Solution: Summary

- Collectively, we do not have strong evidence in favor of the four-class solution (BIC, VLMR)
- Given the mixed results, we could go either way, but we choose to consider the three-class solution with our ultimate analysis in mind
 - Larger sample sizes for comparative purposes
- We proceed with the three-class solution, and now turn to predicting the 2004-2005 outcome indicating any drug use disorders

Modeling the Dependent Variable

- Per Asparouhov and Muthen (2014a, 2014b; see Canvas), trying to model the distal outcome and the latent classes simultaneously has many disadvantages
- Multi-step methods for independent modeling of the distal outcome given information from the LCA have been proposed
- We consider a simple two-step method here, using the **modal predicted class** from our LCA

Step 2: Logistic Regression

- We import the data set saved from Mplus back into Stata (or any other package)
- Then, we regress ANYDUDA on the predicted modal class, using standard logistic regression modeling techniques:

```
logit anyduda ib3.c, or
```

- We could include covariates as well...

Interpretation of Estimates

Logistic regression

Number of obs = 34653

LR chi2(2) = 558.72

Prob > chi2 = 0.0000

Log likelihood = -11018.003

Pseudo R2 = 0.0247

anyduda		Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----							
C							
1		8.071624	.796949	21.15	0.000	6.651473	9.794991
2		7.869833	1.04922	15.47	0.000	6.060128	10.21996
_cons		.1032423	.0019227	-121.93	0.000	.0995419	.1070804

Classes 1 and 2 have substantially increased odds of developing a substance use disorder three years from now, and do not appear to be different from each other. (Would accounting for class prediction uncertainty using a multi-step approach make a difference???)

Would be interesting to examine the same analysis for the four-class solution!

Choosing Software...

- Any of the software discussed last week can be used to implement the “naïve” two-step approach...
 - See the worked R example referenced last week!
- Mplus offers nice graphing capabilities and is especially good at handling complex sampling features
- Mplus makes multi-step modeling approaches, accounting for the uncertainty in class predictions, “automatic”
 - See also Lanza et al. (2013), on Canvas, for implementation in SAS via PROC LCA

Multi-Step Alternatives

- Note that the two-step approach illustrated here does not recognize the possibility of error in the modal classes (i.e., cases did not have 100% probability of falling into their assigned classes)
- **One approach:** impute the class for each person multiple times based on referring random uniform draws to the posterior class probabilities based on the LCA, and perform multiple imputation analysis
- **Other approaches:** see Asparouhov and Muthen (2014a), or Lanza et al. (2013)

“Automatic” Multi-Step Approaches in Mplus

- If you wish to include predictors of latent class membership, include this line in the VARIABLE section of the syntax (**poLCA can also do this**):

```
AUXILIARY = MALE(R3STEP) BLACK(R3STEP) HISPANIC(R3STEP) OTHRACE(R3STEP);
```

- If you wish to predict continuous distal outcomes using your latent classes (Asparouhov and Muthen, 2014b):

```
AUXILIARY = DOUTCOME(BCH);
```

- Nothing “canned” yet in R, but see the presentation on Canvas from Tompsett and De Stavola (2022) about an approach that combines R and Stata to implement these approaches

Thinking Ahead

- These analyses have all ignored the complex sample design features of the NESARC
- Mplus offers the tremendous capability of computing weighted, design-based estimates of these types of complex models
- Next class, we will review the conceptual background of accounting for complex samples when fitting these models, and then revisit this example